

symcoder round-trip demo

electrons=(a,b,c), muons=(p,q)

1 Setup

Particle types: electrons = $\{a, b, c\}$, muons = $\{p, q\}$

Group G: $G = S_{\text{electrons}} \times S_{\text{muons}} = S_3 \times S_2$, order = 12

Operations:

$|\mathbf{x}|$ (rank 1, symmetric)

$\mathbf{x} \cdot \mathbf{y}$ (rank 2, symmetric)

$\mathbf{x}_T \cdot \mathbf{y}_T$ (rank 2, symmetric)

$\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (rank 3, antisymmetric)

Event vectors (3-D):

a: $\mathbf{a} = (1.20, 0.30, 0.50)$

b: $\mathbf{b} = (0.40, 1.10, 0.20)$

c: $\mathbf{c} = (-0.50, 1.20, -0.70)$

p: $\mathbf{p} = (0.00, 0.00, 1.00)$

q: $\mathbf{q} = (0.00, 0.00, -1.00)$

2 Phase 1 Encoding (one orbit per FlavouredOperator in repS)

11 FlavouredOperators in repS

2.1 FlavouredOperator: $|\mathbf{x}|$ flavour = (1, 0)

Canonical representative: $|\mathbf{a}|$

Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $|\mathbf{a}|, |\mathbf{b}|, |\mathbf{c}|$

Eval values: +1.3342, +1.1874, +1.4765

Encoded [3 reals]: +1.1874, +1.3342, +1.4765

2.2 FlavouredOperator: $|\mathbf{x}|$ flavour = (0, 1)

Canonical representative: $|\mathbf{p}|$

Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $|\mathbf{p}|, |\mathbf{q}|$

Eval values: +1.0000, +1.0000
Encoded [2 reals]: +1.0000, +1.0000

2.3 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour = (2, 0)

Canonical representative: $\mathbf{a} \cdot \mathbf{b}$
Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $\mathbf{a} \cdot \mathbf{b}$, $\mathbf{a} \cdot \mathbf{c}$, $\mathbf{b} \cdot \mathbf{c}$
Eval values: +0.9100, -0.5900, +0.9800
Encoded [3 reals]: -0.5900, +0.9100, +0.9800

2.4 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour = (1, 1)

Canonical representative: $\mathbf{a} \cdot \mathbf{p}$
Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $\mathbf{a} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{p}$, $\mathbf{c} \cdot \mathbf{q}$
Eval values: +0.5000, -0.5000, +0.2000, -0.2000, -0.7000, +0.7000
Encoded [6 reals]: -0.7000, -0.5000, -0.2000, +0.2000, +0.5000, +0.7000

2.5 FlavouredOperator: $\mathbf{x} \cdot \mathbf{y}$ flavour = (0, 2)

Canonical representative: $\mathbf{p} \cdot \mathbf{q}$
Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $\mathbf{p} \cdot \mathbf{q}$
Eval values: -1.0000
Encoded [1 reals]: -1.0000

2.6 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour = (2, 0)

Canonical representative: $\mathbf{a}_T \cdot \mathbf{b}_T$
Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $\mathbf{a}_T \cdot \mathbf{b}_T$, $\mathbf{a}_T \cdot \mathbf{c}_T$, $\mathbf{b}_T \cdot \mathbf{c}_T$
Eval values: +0.8100, -0.2400, +1.1200
Encoded [3 reals]: -0.2400, +0.8100, +1.1200

2.7 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour = (1, 1)

Canonical representative: $\mathbf{a}_T \cdot \mathbf{p}_T$
Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $\mathbf{a}_T \cdot \mathbf{p}_T$, $\mathbf{a}_T \cdot \mathbf{q}_T$, $\mathbf{b}_T \cdot \mathbf{p}_T$, $\mathbf{b}_T \cdot \mathbf{q}_T$, $\mathbf{c}_T \cdot \mathbf{p}_T$, $\mathbf{c}_T \cdot \mathbf{q}_T$
Eval values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000
Encoded [6 reals]: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

2.8 FlavouredOperator: $\mathbf{x}_T \cdot \mathbf{y}_T$ flavour = (0, 2)

Canonical representative: $\mathbf{p}_T \cdot \mathbf{q}_T$

Encoder: SortEncoder — stores eval of full orbit

Orbit atoms: $\mathbf{p}_T \cdot \mathbf{q}_T$

Eval values: +0.0000

Encoded [1 reals]: +0.0000

2.9 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour = (3, 0)

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$

Encoder: HalfSortEncoder — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$

represented by : $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$

Eval values: -0.6430

Encoded [1 reals]: +0.6430

2.10 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour = (2, 1)

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$

Encoder: HalfSortEncoder — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $-\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$, $-\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$

represented by : $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$

Eval values: +1.2000, -1.2000, +1.5900, -1.5900, +1.0300, -1.0300

Encoded [6 reals]: +1.0300, +1.0300, +1.2000, +1.2000, +1.5900, +1.5900

2.11 FlavouredOperator: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ flavour = (1, 2)

Canonical representative: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$

Encoder: HalfSortEncoder — stores |eval| of sign= +1 representatives

Orbit atoms: $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$

represented by : $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$

Eval values: +0.0000, +0.0000, +0.0000

Encoded [3 reals]: +0.0000, +0.0000, +0.0000

3 Phase 2 Encoding (OverlapBlocks, complementarity-drop disabled)

3.1 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=2 PairFlavour overlap= (0, 0), type=11_sigma, kind=ASSOC, encoded dim= 2

Encoded values: +2.8284, +2.0000

Atom-pair orbit (multiset; ordering arbitrary):

(|p|, |q|)

(|q|, |p|)

Eval-pair multiset (u, v):

(+1.0000, +1.0000)
(+1.0000, +1.0000)
PairFlavour overlap=(0, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (0, 1),
type=null_self, kind=NULL_SELF, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
(|**p**|, |**p**|)
(|**q**|, |**q**|)
Eval-pair multiset (u, v):
(+1.0000, +1.0000)
(+1.0000, +1.0000)

3.2 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11,
kind=ASSOC, encoded dim= 12
Encoded values: +6.0000, +7.9962, -11.5993, +39.9808, -86.3973, +32.8452, -97.7244, -61.3879, -
6.6542, -76.5400, +15.8009, -14.2916
Atom-pair orbit (multiset; ordering arbitrary):
(|**p**|, |**a**|)
(|**q**|, |**a**|)
(|**p**|, |**b**|)
(|**q**|, |**b**|)
(|**p**|, |**c**|)
(|**q**|, |**c**|)
Eval-pair multiset (u, v):
(+1.0000, +1.4765)
(+1.0000, +1.4765)
(+1.0000, +1.3342)
(+1.0000, +1.3342)
(+1.0000, +1.1874)
(+1.0000, +1.1874)

3.3 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (0, 1), type=11,
kind=ASSOC, encoded dim= 4
Encoded values: +2.0000, +0.0000, +1.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
(|**p**|, $\mathbf{p}_T \cdot \mathbf{q}_T$)
(|**q**|, $\mathbf{p}_T \cdot \mathbf{q}_T$)
Eval-pair multiset (u, v):
(+1.0000, +0.0000)
(+1.0000, +0.0000)

3.4 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=4 **PairFlavour** overlap= (0, 1), type=11,
kind=ASSOC, encoded dim= 4
Encoded values: +2.0000, -2.0000, +0.0000, -2.0000
Atom-pair orbit (multiset; ordering arbitrary):
(|**p**|, $\mathbf{p} \cdot \mathbf{q}$)
(|**q**|, $\mathbf{p} \cdot \mathbf{q}$)

Eval-pair multiset (u, v) :
 (+1.0000, -1.0000)
 (+1.0000, -1.0000)

3.5 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\mathbf{a}_T \cdot \mathbf{q}_T$)
 ($|\mathbf{q}|$, $\mathbf{a}_T \cdot \mathbf{p}_T$)
 ($|\mathbf{p}|$, $\mathbf{b}_T \cdot \mathbf{q}_T$)
 ($|\mathbf{q}|$, $\mathbf{b}_T \cdot \mathbf{p}_T$)
 ($|\mathbf{p}|$, $\mathbf{c}_T \cdot \mathbf{q}_T$)
 ($|\mathbf{q}|$, $\mathbf{c}_T \cdot \mathbf{p}_T$)

Eval-pair multiset (u, v) :

(+1.0000, +0.0000)
 (+1.0000, +0.0000)
 (+1.0000, +0.0000)
 (+1.0000, +0.0000)
 (+1.0000, +0.0000)
 (+1.0000, +0.0000)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\mathbf{a}_T \cdot \mathbf{p}_T$)
 ($|\mathbf{q}|$, $\mathbf{a}_T \cdot \mathbf{q}_T$)
 ($|\mathbf{p}|$, $\mathbf{b}_T \cdot \mathbf{p}_T$)
 ($|\mathbf{q}|$, $\mathbf{b}_T \cdot \mathbf{q}_T$)
 ($|\mathbf{p}|$, $\mathbf{c}_T \cdot \mathbf{p}_T$)
 ($|\mathbf{q}|$, $\mathbf{c}_T \cdot \mathbf{q}_T$)

Eval-pair multiset (u, v) :

(+1.0000, +0.0000)
 (+1.0000, +0.0000)
 (+1.0000, +0.0000)
 (+1.0000, +0.0000)
 (+1.0000, +0.0000)
 (+1.0000, +0.0000)

3.6 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +6.0000, +0.0000, +15.7800, +0.0000, +23.1200, +0.0000, +19.8321, +0.0000, +9.4242, +0.0000, +1.9370, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\mathbf{a} \cdot \mathbf{q}$)
 ($|\mathbf{q}|$, $\mathbf{a} \cdot \mathbf{p}$)

$(|\mathbf{p}|, \mathbf{b} \cdot \mathbf{q})$
 $(|\mathbf{q}|, \mathbf{b} \cdot \mathbf{p})$
 $(|\mathbf{p}|, \mathbf{c} \cdot \mathbf{q})$
 $(|\mathbf{q}|, \mathbf{c} \cdot \mathbf{p})$
 Eval-pair multiset (u, v) :
 $(+1.0000, -0.7000)$
 $(+1.0000, -0.5000)$
 $(+1.0000, -0.2000)$
 $(+1.0000, +0.7000)$
 $(+1.0000, +0.2000)$
 $(+1.0000, +0.5000)$
 PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11,
 kind=ASSOC, encoded dim= 12
Encoded values: +6.0000, +0.0000, +15.7800, +0.0000, +23.1200, +0.0000, +19.8321, +0.0000,
 +9.4242, +0.0000, +1.9370, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 $(|\mathbf{p}|, \mathbf{a} \cdot \mathbf{p})$
 $(|\mathbf{q}|, \mathbf{a} \cdot \mathbf{q})$
 $(|\mathbf{p}|, \mathbf{b} \cdot \mathbf{p})$
 $(|\mathbf{q}|, \mathbf{b} \cdot \mathbf{q})$
 $(|\mathbf{p}|, \mathbf{c} \cdot \mathbf{p})$
 $(|\mathbf{q}|, \mathbf{c} \cdot \mathbf{q})$
 Eval-pair multiset (u, v) :
 $(+1.0000, -0.7000)$
 $(+1.0000, -0.5000)$
 $(+1.0000, -0.2000)$
 $(+1.0000, +0.7000)$
 $(+1.0000, +0.2000)$
 $(+1.0000, +0.5000)$

3.7 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11,
 kind=ASSOC, encoded dim= 12
Encoded values: +6.0000, +3.3800, +11.2559, +16.9000, +5.0236, +32.7347, -8.0034, +30.6042,
 -10.0540, +13.5109, -3.3303, +2.1214
 Atom-pair orbit (multiset; ordering arbitrary):
 $(|\mathbf{p}|, \mathbf{a}_T \cdot \mathbf{b}_T)$
 $(|\mathbf{q}|, \mathbf{a}_T \cdot \mathbf{b}_T)$
 $(|\mathbf{p}|, \mathbf{a}_T \cdot \mathbf{c}_T)$
 $(|\mathbf{q}|, \mathbf{a}_T \cdot \mathbf{c}_T)$
 $(|\mathbf{p}|, \mathbf{b}_T \cdot \mathbf{c}_T)$
 $(|\mathbf{q}|, \mathbf{b}_T \cdot \mathbf{c}_T)$
 Eval-pair multiset (u, v) :
 $(+1.0000, +1.1200)$
 $(+1.0000, +1.1200)$
 $(+1.0000, +0.8100)$
 $(+1.0000, +0.8100)$
 $(+1.0000, -0.2400)$
 $(+1.0000, -0.2400)$

3.8 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +6.0000, +2.6000, +13.7566, +13.0000, +15.0264, +27.6329, +6.2214, +30.8987, -1.6099, +18.1337, -1.8384, +4.4679

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\mathbf{a} \cdot \mathbf{b}$)

($|\mathbf{q}|$, $\mathbf{a} \cdot \mathbf{b}$)

($|\mathbf{p}|$, $\mathbf{a} \cdot \mathbf{c}$)

($|\mathbf{q}|$, $\mathbf{a} \cdot \mathbf{c}$)

($|\mathbf{p}|$, $\mathbf{b} \cdot \mathbf{c}$)

($|\mathbf{q}|$, $\mathbf{b} \cdot \mathbf{c}$)

Eval-pair multiset (u, v) :

(+1.0000, -0.5900)

(+1.0000, -0.5900)

(+1.0000, +0.9800)

(+1.0000, +0.9800)

(+1.0000, +0.9100)

(+1.0000, +0.9100)

3.9 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +6.0000, +0.0000, +15.0000, +0.0000, +20.0000, +0.0000, +15.0000, +0.0000, +6.0000, +0.0000, +1.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{q}|$, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{p}|$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{q}|$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{p}|$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

($|\mathbf{q}|$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

Eval-pair multiset (u, v) :

(+1.0000, +0.0000)

(+1.0000, +0.0000)

(+1.0000, +0.0000)

(+1.0000, -0.0000)

(+1.0000, -0.0000)

(+1.0000, -0.0000)

3.10 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +12.0000, +76.0580, +320.5800, +988.6013, +2326.8903, +4270.6178, +6142.2191, +6877.2694, +5857.9958, +3630.0616, +1483.7636, +314.7569

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{p}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($|\mathbf{q}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($|\mathbf{p}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)

($|\mathbf{q}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)

$(|\mathbf{p}|, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(|\mathbf{q}|, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(|\mathbf{p}|, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(|\mathbf{q}|, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
 $(+1.0000, +1.5900)$
 $(+1.0000, +1.5900)$
 $(+1.0000, +1.2000)$
 $(+1.0000, +1.2000)$
 $(+1.0000, +1.0300)$
 $(+1.0000, +1.0300)$
 $(+1.0000, -1.5900)$
 $(+1.0000, -1.5900)$
... (4 more eval-pairs)
PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=12, kind=ASSOC, encoded dim= 12
Encoded values: +12.0000, +76.0580, +320.5800, +988.6013, +2326.8903, +4270.6178, +6142.2191, +6877.2694, +5857.9958, +3630.0616, +1483.7636, +314.7569
Atom-pair orbit (multiset; ordering arbitrary):
 $(|\mathbf{p}|, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(|\mathbf{q}|, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(|\mathbf{p}|, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(|\mathbf{q}|, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(|\mathbf{p}|, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(|\mathbf{q}|, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(|\mathbf{p}|, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(|\mathbf{q}|, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
 $(+1.0000, +1.5900)$
 $(+1.0000, +1.5900)$
 $(+1.0000, +1.2000)$
 $(+1.0000, +1.2000)$
 $(+1.0000, +1.0300)$
 $(+1.0000, +1.0300)$
 $(+1.0000, -1.5900)$
 $(+1.0000, -1.5900)$
... (4 more eval-pairs)

3.11 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=4 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 4
Encoded values: +4.0000, +6.8269, +5.6538, +1.9978
Atom-pair orbit (multiset; ordering arbitrary):
 $(|\mathbf{p}|, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(|\mathbf{q}|, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(|\mathbf{p}|, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(|\mathbf{q}|, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
Eval-pair multiset (u, v) :
 $(+1.0000, -0.6430)$
 $(+1.0000, -0.6430)$

(+1.0000, +0.6430)
(+1.0000, +0.6430)

3.12 Block: $|\mathbf{x}| \times |\mathbf{x}|$ (block)

PairFlavour overlap=(0, 0) type=11 sigma kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11 sigma, kind=ASSOC, encoded dim= 6

Encoded values: +11.3083, +53.3240, +134.2109, +190.1584, +143.8079, +45.3503

Atom-pair orbit (multiset; ordering arbitrary):

(**|a|**, **|b|**)
(**|a|**, **|c|**)
(**|b|**, **|a|**)
(**|b|**, **|c|**)
(**|c|**, **|a|**)
(**|c|**, **|b|**)

Eval-pair multiset (u, v):

(+1.4765, +1.1874)
(+1.4765, +1.3342)
(+1.3342, +1.1874)
(+1.3342, +1.4765)
(+1.1874, +1.3342)
(+1.1874, +1.4765)

PairFlavour overlap=(1, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (1, 0), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

(**|a|**, **|a|**)
(**|b|**, **|b|**)
(**|c|**, **|c|**)

Eval-pair multiset (u, v):

(+1.1874, +1.1874)
(+1.3342, +1.3342)
(+1.4765, +1.4765)

3.13 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: +3.9981, +0.0000, +5.3073, +0.0000, +2.3391, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

(**|a|**, **$\mathbf{p}_T \cdot \mathbf{q}_T$**)
(**|b|**, **$\mathbf{p}_T \cdot \mathbf{q}_T$**)
(**|c|**, **$\mathbf{p}_T \cdot \mathbf{q}_T$**)

Eval-pair multiset (u, v):

(+1.4765, +0.0000)
(+1.1874, +0.0000)
(+1.3342, +0.0000)

3.14 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: +3.9981, -3.0000, +2.3073, -7.9962, -1.6590, -4.3073

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\mathbf{p} \cdot \mathbf{q}$)

($|\mathbf{b}|$, $\mathbf{p} \cdot \mathbf{q}$)

($|\mathbf{c}|$, $\mathbf{p} \cdot \mathbf{q}$)

Eval-pair multiset (u, v) :

(+1.4765, -1.0000)

(+1.3342, -1.0000)

(+1.1874, -1.0000)

3.15 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +15.9923, +0.0000, +117.1373, +0.0000, +519.6184, +0.0000, +1554.7706, +0.0000, +3305.7787, +0.0000, +5121.4826, +0.0000, +5825.2045, +0.0000, +4827.6985, +0.0000, +2843.1049, +0.0000, +1129.3678, +0.0000, +271.6938, +0.0000, +29.9358, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\mathbf{b}_T \cdot \mathbf{p}_T$)

($|\mathbf{a}|$, $\mathbf{b}_T \cdot \mathbf{q}_T$)

($|\mathbf{a}|$, $\mathbf{c}_T \cdot \mathbf{p}_T$)

($|\mathbf{a}|$, $\mathbf{c}_T \cdot \mathbf{q}_T$)

($|\mathbf{b}|$, $\mathbf{a}_T \cdot \mathbf{p}_T$)

($|\mathbf{b}|$, $\mathbf{a}_T \cdot \mathbf{q}_T$)

($|\mathbf{b}|$, $\mathbf{c}_T \cdot \mathbf{p}_T$)

($|\mathbf{b}|$, $\mathbf{c}_T \cdot \mathbf{q}_T$)

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

(+1.4765, +0.0000)

(+1.4765, +0.0000)

(+1.4765, +0.0000)

(+1.4765, +0.0000)

(+1.3342, +0.0000)

(+1.3342, +0.0000)

(+1.3342, +0.0000)

(+1.3342, +0.0000)

... (4 more eval-pairs)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +7.9962, +0.0000, +26.5993, +0.0000, +47.1165, +0.0000, +46.8716, +0.0000, +24.8287, +0.0000, +5.4714, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\mathbf{a}_T \cdot \mathbf{p}_T$)

($|\mathbf{a}|$, $\mathbf{a}_T \cdot \mathbf{q}_T$)

($|\mathbf{b}|$, $\mathbf{b}_T \cdot \mathbf{p}_T$)

($|\mathbf{b}|$, $\mathbf{b}_T \cdot \mathbf{q}_T$)

($|\mathbf{c}|$, $\mathbf{c}_T \cdot \mathbf{p}_T$)

($|\mathbf{c}|$, $\mathbf{c}_T \cdot \mathbf{q}_T$)

Eval-pair multiset (u, v) :

(+1.4765, +0.0000)

(+1.4765, +0.0000)

(+1.3342, +0.0000)

(+1.3342, +0.0000)

(+1.1874, +0.0000)

(+1.1874, +0.0000)

3.16 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +15.9923, +0.0000, +118.6973, +0.0000, +540.5384, +0.0000, +1681.8576, +0.0000, +3766.3481, +0.0000, +6224.2982, +0.0000, +7648.5883, +0.0000, +6936.4566, +0.0000, +4528.1048, +0.0000, +2020.0109, +0.0000, +553.0411, +0.0000, +70.2940, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\mathbf{b} \cdot \mathbf{p}$)

($|\mathbf{a}|$, $\mathbf{b} \cdot \mathbf{q}$)

($|\mathbf{a}|$, $\mathbf{c} \cdot \mathbf{p}$)

($|\mathbf{a}|$, $\mathbf{c} \cdot \mathbf{q}$)

($|\mathbf{b}|$, $\mathbf{a} \cdot \mathbf{p}$)

($|\mathbf{b}|$, $\mathbf{a} \cdot \mathbf{q}$)

($|\mathbf{b}|$, $\mathbf{c} \cdot \mathbf{p}$)

($|\mathbf{b}|$, $\mathbf{c} \cdot \mathbf{q}$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+1.1874, -0.7000)

(+1.3342, -0.7000)

(+1.4765, -0.5000)

(+1.1874, -0.5000)

(+1.3342, +0.7000)

(+1.1874, +0.7000)

(+1.4765, +0.5000)

(+1.4765, -0.2000)

... (4 more eval-pairs)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +7.9962, +0.0000, +27.3793, +0.0000, +51.1445, +0.0000, +54.8162, +0.0000, +31.8945, +0.0000, +7.8591, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\mathbf{a} \cdot \mathbf{p}$)

($|\mathbf{a}|$, $\mathbf{a} \cdot \mathbf{q}$)

($|\mathbf{b}|$, $\mathbf{b} \cdot \mathbf{p}$)

($|\mathbf{b}|$, $\mathbf{b} \cdot \mathbf{q}$)

($|\mathbf{c}|$, $\mathbf{c} \cdot \mathbf{p}$)

($|\mathbf{c}|$, $\mathbf{c} \cdot \mathbf{q}$)

Eval-pair multiset (u, v):

(+1.4765, -0.7000)

(+1.4765, +0.7000)

(+1.3342, +0.5000)

(+1.1874, +0.2000)

(+1.3342, -0.5000)

(+1.1874, -0.2000)

3.17 Block: $|\mathbf{x}| \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: +3.9981, +1.6900, +4.8633, +4.3515, +1.9181, +2.9918

Atom-pair orbit (multiset; ordering arbitrary):

(|a|, $\mathbf{b}_T \cdot \mathbf{c}_T$)

(|b|, $\mathbf{a}_T \cdot \mathbf{c}_T$)

(|c|, $\mathbf{a}_T \cdot \mathbf{b}_T$)

Eval-pair multiset (u, v) :

(+1.3342, +1.1200)

(+1.4765, +0.8100)

(+1.1874, -0.2400)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +7.9962, +3.3800, +22.8552, +22.6755, +26.7283, +59.6993, +4.7645, +76.6858, -14.1741, +47.4925, -8.2128, +11.0421

Atom-pair orbit (multiset; ordering arbitrary):

(|a|, $\mathbf{a}_T \cdot \mathbf{b}_T$)

(|a|, $\mathbf{a}_T \cdot \mathbf{c}_T$)

(|b|, $\mathbf{a}_T \cdot \mathbf{b}_T$)

(|b|, $\mathbf{b}_T \cdot \mathbf{c}_T$)

(|c|, $\mathbf{a}_T \cdot \mathbf{c}_T$)

(|c|, $\mathbf{b}_T \cdot \mathbf{c}_T$)

Eval-pair multiset (u, v) :

(+1.4765, +1.1200)

(+1.1874, +1.1200)

(+1.1874, +0.8100)

(+1.3342, +0.8100)

(+1.4765, -0.2400)

(+1.3342, -0.2400)

3.18 Block: $|\mathbf{x}| \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: +3.9981, +1.3000, +5.5306, +3.2470, +2.8502, +2.5238

Atom-pair orbit (multiset; ordering arbitrary):

(|a|, $\mathbf{b} \cdot \mathbf{c}$)

(|b|, $\mathbf{a} \cdot \mathbf{c}$)

(|c|, $\mathbf{a} \cdot \mathbf{b}$)

Eval-pair multiset (u, v) :

(+1.4765, +0.9100)

(+1.3342, +0.9800)

(+1.1874, -0.5900)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +7.9962, +2.6000, +25.3559, +17.5430, +39.9914, +48.9122, +30.3174, +69.9165, +6.7367, +51.0380, -2.4527, +15.0917

Atom-pair orbit (multiset; ordering arbitrary):

(|a|, $\mathbf{a} \cdot \mathbf{b}$)

(|a|, $\mathbf{a} \cdot \mathbf{c}$)

(|b|, $\mathbf{a} \cdot \mathbf{b}$)

(|b|, $\mathbf{b} \cdot \mathbf{c}$)

(|c|, $\mathbf{a} \cdot \mathbf{c}$)

(|c|, $\mathbf{b} \cdot \mathbf{c}$)

Eval-pair multiset (u, v) :

(+1.4765, -0.5900)
 (+1.3342, -0.5900)
 (+1.4765, +0.9800)
 (+1.3342, +0.9100)
 (+1.1874, +0.9100)
 (+1.1874, +0.9800)

3.19 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +15.9923, +117.1373, +519.6184, +1554.7706, +3305.7787, +5121.4826, +5825.2045, +4827.6985, +2843.1049, +1129.3678, +271.6938, +29.9358

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{a}|$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{b}|$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{b}|$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{c}|$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{c}|$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{a}|$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{a}|$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+1.4765, +0.0000)
 (+1.4765, +0.0000)
 (+1.4765, +0.0000)
 (+1.4765, -0.0000)
 (+1.3342, -0.0000)
 (+1.3342, -0.0000)
 (+1.3342, +0.0000)
 (+1.3342, +0.0000)

... (4 more eval-pairs)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=12, kind=ASSOC, encoded dim= 6

Encoded values: +7.9962, +26.5993, +47.1165, +46.8716, +24.8287, +5.4714

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{b}|$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{c}|$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{a}|$, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{b}|$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($|\mathbf{c}|$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

Eval-pair multiset (u, v):

(+1.4765, +0.0000)
 (+1.4765, +0.0000)
 (+1.3342, +0.0000)
 (+1.3342, -0.0000)
 (+1.1874, -0.0000)
 (+1.1874, -0.0000)

3.20 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +15.9923, +127.1953, +654.2952, +2406.8990, +6640.4046, +14054.4075, +22965.6441, +28758.7978, +26951.0762, +17999.4163, +7735.4211, +1640.2216

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{a}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)

($|\mathbf{b}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{b}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)

($|\mathbf{c}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($|\mathbf{c}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($|\mathbf{a}|$, $-\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{a}|$, $-\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+1.1874, +1.5900)

(+1.1874, +1.5900)

(+1.4765, +1.2000)

(+1.4765, +1.2000)

(+1.3342, +1.0300)

(+1.3342, +1.0300)

(+1.1874, -1.5900)

(+1.1874, -1.5900)

... (4 more eval-pairs)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +15.9923, +0.0000, +127.1953, +0.0000, +653.3436, +0.0000, +2395.3663, +0.0000, +6572.5381, -0.0000, +13804.9212, -0.0000, +22339.0427, -0.0000, +27649.1661, -0.0000, +25568.2570, +0.0000, +16830.6354, -0.0000, +7126.1835, -0.0000, +1490.5721, -0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{a}|$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)

($|\mathbf{b}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)

($|\mathbf{b}|$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

($|\mathbf{b}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)

($|\mathbf{b}|$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+1.4765, -1.5900)

(+1.3342, -1.5900)

(+1.4765, -1.0300)

(+1.3342, -1.2000)

(+1.1874, -1.2000)

(+1.1874, -1.0300)

(+1.4765, +1.5900)

(+1.3342, +1.5900)

... (4 more eval-pairs)

3.21 Block: $|\mathbf{x}| \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=12, kind=ASSOC, encoded dim= 6

Encoded values: +7.9962, +27.8397, +53.7285, +60.6021, +37.9391, +10.3729

Atom-pair orbit (multiset; ordering arbitrary):

($|\mathbf{a}|, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{b}|, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{c}|, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{a}|, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{b}|, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

($|\mathbf{c}|, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

Eval-pair multiset (u, v):

(+1.4765, +0.6430)

(+1.3342, +0.6430)

(+1.1874, +0.6430)

(+1.4765, -0.6430)

(+1.3342, -0.6430)

(+1.1874, -0.6430)

3.22 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (0, 2), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p}_T \cdot \mathbf{q}_T, \mathbf{p}_T \cdot \mathbf{q}_T$)

Eval-pair multiset (u, v):

(+0.0000, +0.0000)

3.23 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 2) type=11 kind=ASSOC dim=2 **PairFlavour** overlap= (0, 2), type=11, kind=ASSOC, encoded dim= 2

Encoded values: +0.0000, -1.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p}_T \cdot \mathbf{q}_T, \mathbf{p} \cdot \mathbf{q}$)

Eval-pair multiset (u, v):

(+0.0000, -1.0000)

3.24 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p}_T \cdot \mathbf{q}_T, \mathbf{a}_T \cdot \mathbf{p}_T$)

($\mathbf{p}_T \cdot \mathbf{q}_T, \mathbf{a}_T \cdot \mathbf{q}_T$)

($\mathbf{p}_T \cdot \mathbf{q}_T, \mathbf{b}_T \cdot \mathbf{p}_T$)

($\mathbf{p}_T \cdot \mathbf{q}_T, \mathbf{b}_T \cdot \mathbf{q}_T$)

($\mathbf{p}_T \cdot \mathbf{q}_T, \mathbf{c}_T \cdot \mathbf{p}_T$)

($\mathbf{p}_T \cdot \mathbf{q}_T, \mathbf{c}_T \cdot \mathbf{q}_T$)

Eval-pair multiset (u, v) :

(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)

3.25 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +0.0000, +0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, +0.0049, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{a} \cdot \mathbf{p}$)
($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{a} \cdot \mathbf{q}$)
($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{b} \cdot \mathbf{p}$)
($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{b} \cdot \mathbf{q}$)
($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{c} \cdot \mathbf{p}$)
($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{c} \cdot \mathbf{q}$)

Eval-pair multiset (u, v) :

(+0.0000, -0.7000)
(+0.0000, -0.5000)
(+0.0000, +0.7000)
(+0.0000, +0.5000)
(+0.0000, -0.2000)
(+0.0000, +0.2000)

3.26 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: +0.0000, +1.6900, -0.4440, +0.0000, +0.0000, +0.2177

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{a}_T \cdot \mathbf{b}_T$)
($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{a}_T \cdot \mathbf{c}_T$)
($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{b}_T \cdot \mathbf{c}_T$)

Eval-pair multiset (u, v) :

(+0.0000, +1.1200)
(+0.0000, +0.8100)
(+0.0000, -0.2400)

3.27 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: +0.0000, +1.3000, +0.2233, +0.0000, +0.0000, +0.5262

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{a} \cdot \mathbf{b}$)
($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{a} \cdot \mathbf{c}$)
($\mathbf{p}_T \cdot \mathbf{q}_T$, $\mathbf{b} \cdot \mathbf{c}$)

Eval-pair multiset (u, v) :
 (+0.0000, -0.5900)
 (+0.0000, +0.9800)
 (+0.0000, +0.9100)

3.28 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 2), type=12, kind=ASSOC, encoded dim= 6

Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{p}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$

Eval-pair multiset (u, v) :

(+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, -0.0000)
 (+0.0000, -0.0000)
 (+0.0000, -0.0000)

3.29 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{p}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{p}_T \cdot \mathbf{q}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

(+0.0000, -1.5900)
 (+0.0000, -1.5900)
 (+0.0000, -1.2000)
 (+0.0000, -1.2000)
 (+0.0000, -1.0300)
 (+0.0000, -1.0300)
 (+0.0000, +1.5900)
 (+0.0000, +1.5900)

... (4 more eval-pairs)

3.30 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 2

Encoded values: +0.0000, +0.4134

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{p}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$

$(\mathbf{p}_T \cdot \mathbf{q}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$

Eval-pair multiset (u, v) :

(+0.0000, -0.6430)

(+0.0000, +0.6430)

3.31 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (0, 2), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{p} \cdot \mathbf{q})$

Eval-pair multiset (u, v) :

(-1.0000, -1.0000)

3.32 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11, kind=ASSOC, encoded dim= 12

Encoded values: -6.0000, +0.0000, +15.0000, +0.0000, -20.0000, +0.0000, +15.0000, +0.0000, -6.0000, +0.0000, +1.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{a}_T \cdot \mathbf{p}_T)$

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{a}_T \cdot \mathbf{q}_T)$

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{b}_T \cdot \mathbf{p}_T)$

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{b}_T \cdot \mathbf{q}_T)$

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{c}_T \cdot \mathbf{p}_T)$

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{c}_T \cdot \mathbf{q}_T)$

Eval-pair multiset (u, v) :

(-1.0000, +0.0000)

(-1.0000, +0.0000)

(-1.0000, +0.0000)

(-1.0000, +0.0000)

(-1.0000, +0.0000)

(-1.0000, +0.0000)

3.33 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11, kind=ASSOC, encoded dim= 12

Encoded values: -6.0000, +0.0000, +15.7800, +0.0000, -23.1200, +0.0000, +19.8321, +0.0000, -9.4242, +0.0000, +1.9370, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{p})$

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{q})$

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{b} \cdot \mathbf{p})$

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{b} \cdot \mathbf{q})$
 $(\mathbf{p} \cdot \mathbf{q}, \mathbf{c} \cdot \mathbf{p})$
 $(\mathbf{p} \cdot \mathbf{q}, \mathbf{c} \cdot \mathbf{q})$
 Eval-pair multiset (u, v) :
 $(-1.0000, +0.7000)$
 $(-1.0000, +0.5000)$
 $(-1.0000, +0.2000)$
 $(-1.0000, -0.7000)$
 $(-1.0000, -0.2000)$
 $(-1.0000, -0.5000)$

3.34 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: -3.0000, +1.6900, +2.5560, -3.3800, -0.5560, +1.9077

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{a}_T \cdot \mathbf{b}_T)$
 $(\mathbf{p} \cdot \mathbf{q}, \mathbf{a}_T \cdot \mathbf{c}_T)$
 $(\mathbf{p} \cdot \mathbf{q}, \mathbf{b}_T \cdot \mathbf{c}_T)$
 Eval-pair multiset (u, v) :
 $(-1.0000, +1.1200)$
 $(-1.0000, +0.8100)$
 $(-1.0000, -0.2400)$

3.35 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 6

Encoded values: -3.0000, +1.3000, +3.2233, -2.6000, -1.2233, +1.8262

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{p} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{p} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{c})$
 $(\mathbf{p} \cdot \mathbf{q}, \mathbf{b} \cdot \mathbf{c})$
 Eval-pair multiset (u, v) :
 $(-1.0000, -0.5900)$
 $(-1.0000, +0.9800)$
 $(-1.0000, +0.9100)$

3.36 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 2), type=12, kind=ASSOC, encoded dim= 6

Encoded values: -6.0000, +15.0000, -20.0000, +15.0000, -6.0000, +1.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{p} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{p} \cdot \mathbf{q}, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{p} \cdot \mathbf{q}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{p} \cdot \mathbf{q}, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{p} \cdot \mathbf{q}, -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{p} \cdot \mathbf{q}, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 Eval-pair multiset (u, v) :

(-1.0000, +0.0000)
 (-1.0000, +0.0000)
 (-1.0000, +0.0000)
 (-1.0000, -0.0000)
 (-1.0000, -0.0000)
 (-1.0000, -0.0000)

3.37 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=12, kind=ASSOC, encoded dim= 12

Encoded values: -12.0000, +76.0580, -320.5800, +988.6013, -2326.8903, +4270.6178, -6142.2191, +6877.2694, -5857.9958, +3630.0616, -1483.7636, +314.7569

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{p} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\mathbf{p} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(-1.0000, -1.5900)
 (-1.0000, -1.5900)
 (-1.0000, -1.2000)
 (-1.0000, -1.2000)
 (-1.0000, -1.0300)
 (-1.0000, -1.0300)
 (-1.0000, +1.5900)
 (-1.0000, +1.5900)

... (4 more eval-pairs)

3.38 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=2 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 2

Encoded values: -2.0000, +1.4134

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{p} \cdot \mathbf{q}$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{p} \cdot \mathbf{q}$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

Eval-pair multiset (u, v):

(-1.0000, -0.6430)
 (-1.0000, +0.6430)

3.39 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11_sigma, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{b}_T \cdot \mathbf{q}_T)$

$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{b}_T \cdot \mathbf{p}_T)$

$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{c}_T \cdot \mathbf{q}_T)$

$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{c}_T \cdot \mathbf{p}_T)$

$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{a}_T \cdot \mathbf{q}_T)$

$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{a}_T \cdot \mathbf{p}_T)$

$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{c}_T \cdot \mathbf{q}_T)$

$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{c}_T \cdot \mathbf{p}_T)$

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

... (4 more eval-pairs)

PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11_sigma, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{b}_T \cdot \mathbf{p}_T)$

$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{b}_T \cdot \mathbf{q}_T)$

$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{c}_T \cdot \mathbf{p}_T)$

$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{c}_T \cdot \mathbf{q}_T)$

$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{a}_T \cdot \mathbf{p}_T)$

$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{a}_T \cdot \mathbf{q}_T)$

$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{c}_T \cdot \mathbf{p}_T)$

$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{c}_T \cdot \mathbf{q}_T)$

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

$(+0.0000, +0.0000)$

... (4 more eval-pairs)

PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=11_sigma, kind=ASSOC, encoded dim= 6

Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{a}_T \cdot \mathbf{q}_T)$

$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{a}_T \cdot \mathbf{p}_T)$

$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{b}_T \cdot \mathbf{q}_T)$

$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{b}_T \cdot \mathbf{p}_T)$

$(\mathbf{c}_T \cdot \mathbf{p}_T, \mathbf{c}_T \cdot \mathbf{q}_T)$

$(\mathbf{c}_T \cdot \mathbf{q}_T, \mathbf{c}_T \cdot \mathbf{p}_T)$
 Eval-pair multiset (u, v) :
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (1, 1),
 type=null_self, kind=NULL_SELF, encoded dim= 0
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{a}_T \cdot \mathbf{p}_T)$
 $(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{a}_T \cdot \mathbf{q}_T)$
 $(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{b}_T \cdot \mathbf{p}_T)$
 $(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{b}_T \cdot \mathbf{q}_T)$
 $(\mathbf{c}_T \cdot \mathbf{p}_T, \mathbf{c}_T \cdot \mathbf{p}_T)$
 $(\mathbf{c}_T \cdot \mathbf{q}_T, \mathbf{c}_T \cdot \mathbf{q}_T)$
 Eval-pair multiset (u, v) :
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)

3.40 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (0, 0), type=11,
 kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +1.5600, +0.0000, +0.0000, +0.0000, +0.9126, +0.0000, +0.0000,
 +0.0000, +0.2471, +0.0000, +0.0000, +0.0000, +0.0308, +0.0000, +0.0000, +0.0000, +0.0015, +0.0000,
 +0.0000, +0.0000, +0.0000, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{b} \cdot \mathbf{q})$
 $(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{b} \cdot \mathbf{p})$
 $(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{c} \cdot \mathbf{q})$
 $(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{c} \cdot \mathbf{p})$
 $(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{a} \cdot \mathbf{q})$
 $(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{a} \cdot \mathbf{p})$
 $(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{c} \cdot \mathbf{q})$
 $(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{c} \cdot \mathbf{p})$
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 (+0.0000, +0.7000)
 (+0.0000, +0.7000)
 (+0.0000, -0.7000)
 (+0.0000, -0.7000)
 (+0.0000, -0.5000)
 (+0.0000, -0.5000)
 (+0.0000, +0.5000)
 (+0.0000, +0.5000)
 ... (4 more eval-pairs)
 PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (0, 1), type=11,

kind=ASSOC, encoded dim= 24

Encoded values: +0.0000, +0.0000, +1.5600, +0.0000, +0.0000, +0.0000, +0.9126, +0.0000, +0.0000, +0.0000, +0.2471, +0.0000, +0.0000, +0.0000, +0.0308, +0.0000, +0.0000, +0.0000, +0.0015, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{b} \cdot \mathbf{p})$

$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{b} \cdot \mathbf{q})$

$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{c} \cdot \mathbf{p})$

$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{c} \cdot \mathbf{q})$

$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{a} \cdot \mathbf{p})$

$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{a} \cdot \mathbf{q})$

$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{c} \cdot \mathbf{p})$

$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{c} \cdot \mathbf{q})$

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

(+0.0000, +0.7000)

(+0.0000, +0.7000)

(+0.0000, -0.7000)

(+0.0000, -0.7000)

(+0.0000, -0.5000)

(+0.0000, -0.5000)

(+0.0000, +0.5000)

(+0.0000, +0.5000)

... (4 more eval-pairs)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1,0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +0.0000, +0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, +0.0049, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{a} \cdot \mathbf{q})$

$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{a} \cdot \mathbf{p})$

$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{b} \cdot \mathbf{q})$

$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{b} \cdot \mathbf{p})$

$(\mathbf{c}_T \cdot \mathbf{p}_T, \mathbf{c} \cdot \mathbf{q})$

$(\mathbf{c}_T \cdot \mathbf{q}_T, \mathbf{c} \cdot \mathbf{p})$

Eval-pair multiset (u, v) :

(+0.0000, -0.7000)

(+0.0000, -0.5000)

(+0.0000, +0.7000)

(+0.0000, +0.5000)

(+0.0000, -0.2000)

(+0.0000, +0.2000)

PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1,1), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +0.0000, +0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000, +0.0000, +0.0049, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{a} \cdot \mathbf{p})$

$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{a} \cdot \mathbf{q})$

$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{b} \cdot \mathbf{p})$

$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{b} \cdot \mathbf{q})$

$(\mathbf{c}_T \cdot \mathbf{p}_T, \mathbf{c} \cdot \mathbf{p})$

$(\mathbf{c}_T \cdot \mathbf{q}_T, \mathbf{c} \cdot \mathbf{q})$

Eval-pair multiset (u, v) :

(+0.0000, -0.7000)
(+0.0000, -0.5000)
(+0.0000, +0.7000)
(+0.0000, +0.5000)
(+0.0000, -0.2000)
(+0.0000, +0.2000)

3.41 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +3.3800, -3.7441, +0.0000, +0.0000, -1.0653, -0.5388, +0.0000, +0.0000, -0.1933, -0.0474, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a}_T \cdot \mathbf{p}_T$, $\mathbf{b}_T \cdot \mathbf{c}_T$)
($\mathbf{a}_T \cdot \mathbf{q}_T$, $\mathbf{b}_T \cdot \mathbf{c}_T$)
($\mathbf{b}_T \cdot \mathbf{p}_T$, $\mathbf{a}_T \cdot \mathbf{c}_T$)
($\mathbf{b}_T \cdot \mathbf{q}_T$, $\mathbf{a}_T \cdot \mathbf{c}_T$)
($\mathbf{c}_T \cdot \mathbf{p}_T$, $\mathbf{a}_T \cdot \mathbf{b}_T$)
($\mathbf{c}_T \cdot \mathbf{q}_T$, $\mathbf{a}_T \cdot \mathbf{b}_T$)

Eval-pair multiset (u, v) :

(+0.0000, +1.1200)
(+0.0000, +1.1200)
(+0.0000, +0.8100)
(+0.0000, +0.8100)
(+0.0000, -0.2400)
(+0.0000, -0.2400)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +0.0000, +6.7600, -18.9126, +0.0000, +0.0000, -27.4406, +20.1419, +0.0000, +0.0000, +3.9480, +4.1119, +0.0000, +0.0000, +2.2752, +0.2333, +0.0000, +0.0000, +0.3093, +0.0137, +0.0000, +0.0000, +0.0183, +0.0022, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a}_T \cdot \mathbf{p}_T$, $\mathbf{a}_T \cdot \mathbf{b}_T$)
($\mathbf{a}_T \cdot \mathbf{q}_T$, $\mathbf{a}_T \cdot \mathbf{b}_T$)
($\mathbf{a}_T \cdot \mathbf{p}_T$, $\mathbf{a}_T \cdot \mathbf{c}_T$)
($\mathbf{a}_T \cdot \mathbf{q}_T$, $\mathbf{a}_T \cdot \mathbf{c}_T$)
($\mathbf{b}_T \cdot \mathbf{p}_T$, $\mathbf{a}_T \cdot \mathbf{b}_T$)
($\mathbf{b}_T \cdot \mathbf{q}_T$, $\mathbf{a}_T \cdot \mathbf{b}_T$)
($\mathbf{b}_T \cdot \mathbf{p}_T$, $\mathbf{b}_T \cdot \mathbf{c}_T$)
($\mathbf{b}_T \cdot \mathbf{q}_T$, $\mathbf{b}_T \cdot \mathbf{c}_T$)

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

(+0.0000, +1.1200)
(+0.0000, +1.1200)
(+0.0000, +1.1200)
(+0.0000, +1.1200)
(+0.0000, +0.8100)
(+0.0000, +0.8100)
(+0.0000, +0.8100)
(+0.0000, +0.8100)

... (4 more eval-pairs)

3.42 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +2.6000, -1.2434, +0.0000, +0.0000, +1.6329, -1.3182, +0.0000, +0.0000, +0.2350, -0.2768, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{b} \cdot \mathbf{c})$$
$$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{b} \cdot \mathbf{c})$$
$$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{a} \cdot \mathbf{c})$$
$$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{a} \cdot \mathbf{c})$$
$$(\mathbf{c}_T \cdot \mathbf{p}_T, \mathbf{a} \cdot \mathbf{b})$$
$$(\mathbf{c}_T \cdot \mathbf{q}_T, \mathbf{a} \cdot \mathbf{b})$$

Eval-pair multiset (u, v) :

 $(+0.0000, -0.5900)$
$$(+0.0000, -0.5900)$$
 $(+0.0000, +0.9800)$ $(+0.0000, +0.9800)$ $(+0.0000, +0.9100)$ $(+0.0000, +0.9100)$ PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +0.0000, +5.2000, -9.2468, +0.0000, +0.0000, -3.1999, -9.5814, +0.0000, +0.0000, -10.4452, -1.1640, +0.0000, +0.0000, -6.3288, +1.6586, +0.0000, +0.0000, -1.5236, +0.6746, +0.0000, +0.0000, -0.1301, +0.0766, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{a} \cdot \mathbf{b})$$
$$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{a} \cdot \mathbf{b})$$
$$(\mathbf{a}_T \cdot \mathbf{p}_T, \mathbf{a} \cdot \mathbf{c})$$
$$(\mathbf{a}_T \cdot \mathbf{q}_T, \mathbf{a} \cdot \mathbf{c})$$
$$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{a} \cdot \mathbf{b})$$
$$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{a} \cdot \mathbf{b})$$
$$(\mathbf{b}_T \cdot \mathbf{p}_T, \mathbf{b} \cdot \mathbf{c})$$
$$(\mathbf{b}_T \cdot \mathbf{q}_T, \mathbf{b} \cdot \mathbf{c})$$

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

 $(+0.0000, -0.5900)$ $(+0.0000, -0.5900)$
$$(+0.0000, -0.5900)$$
 $(+0.0000, -0.5900)$ $(+0.0000, +0.9800)$ $(+0.0000, +0.9800)$ $(+0.0000, +0.9800)$ $(+0.0000, +0.9800)$

... (4 more eval-pairs)

3.43 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (0, 1), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000,

+0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{a}_T \cdot \mathbf{q}_T$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{a}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{a}_T \cdot \mathbf{q}_T$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{b}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{b}_T \cdot \mathbf{q}_T$, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{b}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{b}_T \cdot \mathbf{q}_T$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
 ... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, -0.0000)
 (+0.0000, -0.0000)
 (+0.0000, -0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 ... (4 more eval-pairs)

PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 1), type=11, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{a}_T \cdot \mathbf{q}_T$, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{b}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{b}_T \cdot \mathbf{q}_T$, $-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{c}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
 ($\mathbf{c}_T \cdot \mathbf{q}_T$, $-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)

Eval-pair multiset (u, v):

(+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, +0.0000)
 (+0.0000, -0.0000)
 (+0.0000, -0.0000)
 (+0.0000, -0.0000)

3.44 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{a}_T \cdot \mathbf{q}_T$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{b}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{b}_T \cdot \mathbf{q}_T$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{c}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

$(\mathbf{c}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{a}_T \cdot \mathbf{p}_T, -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{a}_T \cdot \mathbf{q}_T, -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
 $(+0.0000, -1.5900)$
 $(+0.0000, -1.5900)$
 $(+0.0000, -1.2000)$
 $(+0.0000, -1.2000)$
 $(+0.0000, -1.0300)$
 $(+0.0000, -1.0300)$
 $(+0.0000, +1.5900)$
 $(+0.0000, +1.5900)$
... (4 more eval-pairs)
PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=12, kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163
Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a}_T \cdot \mathbf{p}_T, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{a}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{b}_T \cdot \mathbf{p}_T, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{b}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{c}_T \cdot \mathbf{p}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{c}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{a}_T \cdot \mathbf{p}_T, -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{a}_T \cdot \mathbf{q}_T, -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
 $(+0.0000, -1.5900)$
 $(+0.0000, -1.5900)$
 $(+0.0000, -1.2000)$
 $(+0.0000, -1.2000)$
 $(+0.0000, -1.0300)$
 $(+0.0000, -1.0300)$
 $(+0.0000, +1.5900)$
 $(+0.0000, +1.5900)$
... (4 more eval-pairs)
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +10.0580, +0.0000, +0.0000, +0.0000, +40.9913, +0.0000, +0.0000, +0.0000, +86.6819, +0.0000, +0.0000, +0.0000, +100.4717, +0.0000, +0.0000, +0.0000, +60.6378, +0.0000, +0.0000, +0.0000, +14.9163, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a}_T \cdot \mathbf{p}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{a}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{a}_T \cdot \mathbf{p}_T, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{a}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{b}_T \cdot \mathbf{p}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{b}_T \cdot \mathbf{q}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{b}_T \cdot \mathbf{p}_T, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{b}_T \cdot \mathbf{q}_T, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
... (4 more atom-pairs)

Eval-pair multiset (u, v) :
 (+0.0000, -1.5900)
 (+0.0000, -1.5900)
 (+0.0000, -1.2000)
 (+0.0000, -1.2000)
 (+0.0000, -1.0300)
 (+0.0000, -1.0300)
 (+0.0000, +1.5900)
 (+0.0000, +1.5900)
 ... (4 more eval-pairs)
 PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 1), type=11,
 kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +10.0580, +0.0000, +0.0000, +0.0000, +40.9913, +0.0000,
 +0.0000, +0.0000, +86.6819, +0.0000, +0.0000, +0.0000, +100.4717, +0.0000, +0.0000, +0.0000,
 +60.6378, +0.0000, +0.0000, +0.0000, +14.9163, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 ($\mathbf{a}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\mathbf{a}_T \cdot \mathbf{q}_T$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\mathbf{a}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{a}_T \cdot \mathbf{q}_T$, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 ($\mathbf{b}_T \cdot \mathbf{p}_T$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 ($\mathbf{b}_T \cdot \mathbf{q}_T$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ($\mathbf{b}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 ($\mathbf{b}_T \cdot \mathbf{q}_T$, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 (+0.0000, -1.5900)
 (+0.0000, -1.5900)
 (+0.0000, -1.2000)
 (+0.0000, -1.2000)
 (+0.0000, -1.0300)
 (+0.0000, -1.0300)
 (+0.0000, +1.5900)
 (+0.0000, +1.5900)
 ... (4 more eval-pairs)

3.45 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=12,
 kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +2.4807, +0.0000, +2.5641, +0.0000, +1.4135, +0.0000, +0.4383, +0.0000,
 +0.0725, +0.0000, +0.0050
 Atom-pair orbit (multiset; ordering arbitrary):
 ($\mathbf{a}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{a}_T \cdot \mathbf{q}_T$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{b}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{b}_T \cdot \mathbf{q}_T$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{c}_T \cdot \mathbf{p}_T$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{c}_T \cdot \mathbf{q}_T$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{a}_T \cdot \mathbf{p}_T$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\mathbf{a}_T \cdot \mathbf{q}_T$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :

(+0.0000, -0.6430)
 (+0.0000, -0.6430)
 (+0.0000, -0.6430)
 (+0.0000, -0.6430)
 (+0.0000, -0.6430)
 (+0.0000, -0.6430)
 (+0.0000, +0.6430)
 (+0.0000, +0.6430)
 ... (4 more eval-pairs)

3.46 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11_sigma, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, -1.5600, +0.0000, +0.9126, -0.0000, -0.2198, +0.0000, +0.0095, +0.0000, +0.0027, +0.0000, +0.0129

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{q}$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{p}$)
 ($\mathbf{a} \cdot \mathbf{p}$, $\mathbf{c} \cdot \mathbf{q}$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{p}$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{q}$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{p}$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\mathbf{c} \cdot \mathbf{q}$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{p}$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+0.5000, +0.7000)
 (+0.7000, +0.5000)
 (+0.7000, +0.2000)
 (+0.2000, +0.7000)
 (+0.5000, -0.2000)
 (+0.2000, -0.5000)
 (-0.2000, -0.7000)
 (-0.5000, -0.7000)

... (4 more eval-pairs)

PairFlavour overlap=(0, 1) type=11_sigma kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=11_sigma, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +1.5600, -0.0000, +0.9126, -0.0000, +0.2198, -0.0000, +0.0095, -0.0000, -0.0027, +0.0000, +0.0129

Atom-pair orbit (multiset; ordering arbitrary):

($\mathbf{a} \cdot \mathbf{p}$, $\mathbf{b} \cdot \mathbf{p}$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\mathbf{b} \cdot \mathbf{q}$)
 ($\mathbf{a} \cdot \mathbf{p}$, $\mathbf{c} \cdot \mathbf{p}$)
 ($\mathbf{a} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{q}$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\mathbf{a} \cdot \mathbf{p}$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\mathbf{a} \cdot \mathbf{q}$)
 ($\mathbf{b} \cdot \mathbf{p}$, $\mathbf{c} \cdot \mathbf{p}$)
 ($\mathbf{b} \cdot \mathbf{q}$, $\mathbf{c} \cdot \mathbf{q}$)

... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+0.7000, -0.5000)
 (+0.5000, -0.7000)

(+0.2000, -0.7000)
 (+0.7000, -0.2000)
 (-0.2000, -0.5000)
 (-0.5000, -0.2000)
 (-0.7000, +0.2000)
 (-0.7000, +0.5000)
 ... (4 more eval-pairs)
 PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=11_sigma,
 kind=ASSOC, encoded dim= 6
Encoded values: +0.0000, +1.5600, +0.0000, +0.6084, +0.0000, +0.0392
 Atom-pair orbit (multiset; ordering arbitrary):
 (**a** · **p**, **a** · **q**)
 (**a** · **q**, **a** · **p**)
 (**b** · **p**, **b** · **q**)
 (**b** · **q**, **b** · **p**)
 (**c** · **p**, **c** · **q**)
 (**c** · **q**, **c** · **p**)
 Eval-pair multiset (u, v):
 (+0.7000, -0.7000)
 (+0.5000, -0.5000)
 (-0.7000, +0.7000)
 (-0.5000, +0.5000)
 (+0.2000, -0.2000)
 (-0.2000, +0.2000)
 PairFlavour overlap=(1, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (1, 1),
 type=null_self, kind=NULL_SELF, encoded dim= 0
 Atom-pair orbit (multiset; ordering arbitrary):
 (**a** · **p**, **a** · **p**)
 (**a** · **q**, **a** · **q**)
 (**b** · **p**, **b** · **p**)
 (**b** · **q**, **b** · **q**)
 (**c** · **p**, **c** · **p**)
 (**c** · **q**, **c** · **q**)
 Eval-pair multiset (u, v):
 (-0.7000, -0.7000)
 (-0.5000, -0.5000)
 (-0.2000, -0.2000)
 (+0.2000, +0.2000)
 (+0.5000, +0.5000)
 (+0.7000, +0.7000)

3.47 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11,
 kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +3.3800, -4.5241, -0.0000, +0.0000, -2.3671, -0.0651, +0.0000, -0.0000,
 -0.3392, -0.1683, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 (**a** · **p**, **b_T** · **c_T**)
 (**a** · **q**, **b_T** · **c_T**)
 (**b** · **p**, **a_T** · **c_T**)
 (**b** · **q**, **a_T** · **c_T**)
 (**c** · **p**, **a_T** · **b_T**)

$(\mathbf{c} \cdot \mathbf{q}, \mathbf{a}_T \cdot \mathbf{b}_T)$
 Eval-pair multiset (u, v) :
 (+0.5000, +1.1200)
 (+0.7000, +0.8100)
 (-0.5000, +1.1200)
 (-0.7000, +0.8100)
 (+0.2000, -0.2400)
 (-0.2000, -0.2400)
 PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11,
 kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +6.7600, -20.4726, +0.0000, -0.0000, -36.6844, +44.4512, -0.0000, +0.0000,
 +41.4253, -33.6972, -0.0000, +0.0000, -24.2761, +14.0721, +0.0000, -0.0000, +6.4766, -2.8173, -0.0000,
 +0.0000, -1.1257, +0.2399, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \mathbf{a}_T \cdot \mathbf{b}_T)$
 $(\mathbf{a} \cdot \mathbf{q}, \mathbf{a}_T \cdot \mathbf{b}_T)$
 $(\mathbf{a} \cdot \mathbf{p}, \mathbf{a}_T \cdot \mathbf{c}_T)$
 $(\mathbf{a} \cdot \mathbf{q}, \mathbf{a}_T \cdot \mathbf{c}_T)$
 $(\mathbf{b} \cdot \mathbf{p}, \mathbf{a}_T \cdot \mathbf{b}_T)$
 $(\mathbf{b} \cdot \mathbf{q}, \mathbf{a}_T \cdot \mathbf{b}_T)$
 $(\mathbf{b} \cdot \mathbf{p}, \mathbf{b}_T \cdot \mathbf{c}_T)$
 $(\mathbf{b} \cdot \mathbf{q}, \mathbf{b}_T \cdot \mathbf{c}_T)$
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 (-0.7000, -0.2400)
 (-0.7000, +1.1200)
 (-0.5000, -0.2400)
 (+0.7000, -0.2400)
 (+0.5000, -0.2400)
 (+0.7000, +1.1200)
 (-0.5000, +0.8100)
 (-0.2000, +1.1200)
 ... (4 more eval-pairs)

3.48 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(0, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=11,
 kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +2.6000, -2.0234, +0.0000, +0.0000, +0.9395, -1.6868, +0.0000, +0.0000,
 -0.0250, -0.6192, +0.0000
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \mathbf{b} \cdot \mathbf{c})$
 $(\mathbf{a} \cdot \mathbf{q}, \mathbf{b} \cdot \mathbf{c})$
 $(\mathbf{b} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{c})$
 $(\mathbf{b} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{c})$
 $(\mathbf{c} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{c} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{b})$
 Eval-pair multiset (u, v) :
 (-0.7000, +0.9100)
 (-0.5000, +0.9800)
 (+0.7000, +0.9100)
 (+0.5000, +0.9800)
 (-0.2000, -0.5900)

(+0.2000, -0.5900)
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11,
kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +5.2000, -10.8068, -0.0000, +0.0000, -10.6185, +5.1708, +0.0000, -0.0000,
+5.4759, -11.3264, +0.0000, +0.0000, -10.0996, +2.7457, +0.0000, +0.0000, +0.6239, -2.8179, +0.0000,
-0.0000, -2.5284, +0.6807, -0.0000
Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{a} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{a} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{c})$
 $(\mathbf{a} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{c})$
 $(\mathbf{b} \cdot \mathbf{p}, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{b} \cdot \mathbf{q}, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{b} \cdot \mathbf{p}, \mathbf{b} \cdot \mathbf{c})$
 $(\mathbf{b} \cdot \mathbf{q}, \mathbf{b} \cdot \mathbf{c})$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
(-0.7000, -0.5900)
(-0.5000, -0.5900)
(+0.7000, -0.5900)
(+0.5000, -0.5900)
(-0.7000, +0.9800)
(+0.7000, +0.9800)
(-0.5000, +0.9100)
(+0.5000, +0.9100)
... (4 more eval-pairs)

3.49 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (0, 1), type=11,
kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, -1.5600, +0.0000, +0.0000, +0.0000, +0.9126, +0.0000, +0.0000,
+0.0000, -0.2471, +0.0000, -0.0000, +0.0000, +0.0308, +0.0000, +0.0000, +0.0000, -0.0015, +0.0000,
-0.0000, +0.0000, +0.0000, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{q}, -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{q}, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{q}, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{q}, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
(+0.7000, +0.0000)
(+0.7000, +0.0000)
(-0.7000, +0.0000)
(-0.7000, -0.0000)
(-0.5000, -0.0000)
(-0.5000, -0.0000)
(+0.5000, +0.0000)
(+0.5000, +0.0000)

... (4 more eval-pairs)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 1), type=11,
kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +0.0000, -0.7800, +0.0000, +0.0000, +0.0000, +0.1521, +0.0000, +0.0000,
+0.0000, -0.0049, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{q}, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{q}, -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{c} \cdot \mathbf{p}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{c} \cdot \mathbf{q}, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
Eval-pair multiset (u, v) :
(-0.7000, +0.0000)
(-0.5000, +0.0000)
(+0.7000, +0.0000)
(+0.5000, -0.0000)
(-0.2000, -0.0000)
(+0.2000, -0.0000)

3.50 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 0), type=12,
kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +8.4980, +0.0000, +32.6451, +0.0000, +74.9539, +0.0000, +108.9323,
+0.0000, +93.2282, +0.0000, +42.2161
Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{q}, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{c} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{c} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{a} \cdot \mathbf{p}, -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{q}, -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
(+0.2000, -1.5900)
(-0.2000, -1.5900)
(+0.7000, -1.2000)
(+0.5000, -1.0300)
(-0.7000, -1.2000)
(-0.5000, -1.0300)
(+0.7000, +1.2000)
(+0.2000, +1.5900)
... (4 more eval-pairs)
PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=12,
kind=ASSOC, encoded dim= 12
Encoded values: +0.0000, +8.4980, +0.0000, +32.6451, +0.0000, +74.9539, +0.0000, +108.9323,
+0.0000, +93.2282, +0.0000, +42.2161
Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{a} \cdot \mathbf{q}, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$

$(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{b} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{c} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{c} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{p}, -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{a} \cdot \mathbf{q}, -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
(+0.2000, -1.5900)
(-0.2000, -1.5900)
(+0.7000, -1.2000)
(+0.5000, -1.0300)
(-0.7000, -1.2000)
(-0.5000, -1.0300)
(+0.7000, +1.2000)
(+0.2000, +1.5900)
... (4 more eval-pairs)
PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11,
kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +8.4980, +6.3900, -0.0000, -0.0000, +14.3494, +44.1434, -
0.0000, +0.0000, -28.3068, +104.9126, -0.0000, -0.0000, -107.5651, +100.7828, +0.0000, -0.0000, -
108.4640, +22.6706, +0.0000, -0.0000, -33.8292, -11.7340
Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{b} \cdot \mathbf{p}, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{q}, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{q}, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
(+0.7000, -1.5900)
(+0.5000, -1.5900)
(-0.2000, -1.2000)
(+0.7000, -1.0300)
(+0.5000, -1.2000)
(+0.2000, -1.0300)
(-0.7000, +1.5900)
(-0.5000, +1.5900)
... (4 more eval-pairs)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 1), type=11,
kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +8.4980, -6.3900, +0.0000, +0.0000, +14.3494, -44.1434, +0.0000,
-0.0000, -28.3068, -104.9126, +0.0000, +0.0000, -107.5651, -100.7828, +0.0000, +0.0000, -108.4640, -
22.6706, +0.0000, -0.0000, -33.8292, +11.7340
Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{a} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{a} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{p}, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$

$(\mathbf{b} \cdot \mathbf{q}, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{b} \cdot \mathbf{q}, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 $(+0.7000, +1.5900)$
 $(+0.5000, +1.5900)$
 $(-0.2000, +1.2000)$
 $(+0.7000, +1.0300)$
 $(+0.5000, +1.2000)$
 $(+0.2000, +1.0300)$
 $(-0.7000, -1.5900)$
 $(-0.5000, -1.5900)$
 ... (4 more eval-pairs)

3.51 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=12,
 kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +0.9207, -0.0000, +1.5418, -0.0000, +1.1362, -0.0000, +0.7563, -0.0000,
 +0.3093, -0.0000, +0.0739

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{a} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{b} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{b} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{c} \cdot \mathbf{p}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{c} \cdot \mathbf{q}, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{a} \cdot \mathbf{p}, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{a} \cdot \mathbf{q}, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 $(+0.7000, -0.6430)$
 $(-0.7000, -0.6430)$
 $(+0.5000, -0.6430)$
 $(-0.5000, -0.6430)$
 $(+0.2000, -0.6430)$
 $(-0.2000, -0.6430)$
 $(+0.7000, +0.6430)$
 $(-0.7000, +0.6430)$
 ... (4 more eval-pairs)

3.52 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x}_T \cdot \mathbf{y}_T$ (block)

PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=11_sigma,
 kind=ASSOC, encoded dim= 6

Encoded values: +4.7800, +10.5364, +14.1307, +12.0340, +6.2732, +1.7889

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{b}_T, \mathbf{a}_T \cdot \mathbf{c}_T)$
 $(\mathbf{a}_T \cdot \mathbf{c}_T, \mathbf{a}_T \cdot \mathbf{b}_T)$
 $(\mathbf{a}_T \cdot \mathbf{b}_T, \mathbf{b}_T \cdot \mathbf{c}_T)$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, \mathbf{a}_T \cdot \mathbf{b}_T)$

$(\mathbf{a}_T \cdot \mathbf{c}_T, \mathbf{b}_T \cdot \mathbf{c}_T)$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, \mathbf{a}_T \cdot \mathbf{c}_T)$
 Eval-pair multiset (u, v) :
 $(-0.2400, +1.1200)$
 $(-0.2400, +0.8100)$
 $(+0.8100, +1.1200)$
 $(+1.1200, +0.8100)$
 $(+1.1200, -0.2400)$
 $(+0.8100, -0.2400)$
 PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (2, 0),
 type=null_self, kind=NULL_SELF, encoded dim= 0
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a}_T \cdot \mathbf{b}_T, \mathbf{a}_T \cdot \mathbf{b}_T)$
 $(\mathbf{a}_T \cdot \mathbf{c}_T, \mathbf{a}_T \cdot \mathbf{c}_T)$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, \mathbf{b}_T \cdot \mathbf{c}_T)$
 Eval-pair multiset (u, v) :
 $(-0.2400, -0.2400)$
 $(+0.8100, +0.8100)$
 $(+1.1200, +1.1200)$

3.53 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=11,
 kind=ASSOC, encoded dim= 12
Encoded values: +3.3800, +2.6000, +2.5007, +8.5673, -4.5116, +11.3877, -9.5180, +5.4343, -6.3584,
 -0.9672, -1.2926, -1.7926
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a}_T \cdot \mathbf{b}_T, \mathbf{a} \cdot \mathbf{c})$
 $(\mathbf{a}_T \cdot \mathbf{c}_T, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{a}_T \cdot \mathbf{b}_T, \mathbf{b} \cdot \mathbf{c})$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{a}_T \cdot \mathbf{c}_T, \mathbf{b} \cdot \mathbf{c})$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, \mathbf{a} \cdot \mathbf{c})$
 Eval-pair multiset (u, v) :
 $(+1.1200, -0.5900)$
 $(+0.8100, -0.5900)$
 $(+1.1200, +0.9100)$
 $(+0.8100, +0.9800)$
 $(-0.2400, +0.9800)$
 $(-0.2400, +0.9100)$
 PairFlavour overlap=(2, 0) type=11 kind=ASSOC dim=6 **PairFlavour** overlap= (2, 0), type=11,
 kind=ASSOC, encoded dim= 6
Encoded values: +1.6900, +1.3000, +0.6673, +0.2207, +1.0660, -0.4442
 Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a}_T \cdot \mathbf{b}_T, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{a}_T \cdot \mathbf{c}_T, \mathbf{a} \cdot \mathbf{c})$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, \mathbf{b} \cdot \mathbf{c})$
 Eval-pair multiset (u, v) :
 $(+1.1200, +0.9800)$
 $(+0.8100, +0.9100)$
 $(-0.2400, -0.5900)$

3.54 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (0,0), type=12, kind=ASSOC, encoded dim= 6

Encoded values: +3.3800, +3.7441, +1.0653, -0.5388, -0.1933, +0.0474

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{b}_T, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{a}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{b}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{a}_T \cdot \mathbf{b}_T, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{a}_T \cdot \mathbf{c}_T, -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{b}_T \cdot \mathbf{c}_T, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$

Eval-pair multiset (u, v) :

(+1.1200, +0.0000)

(+1.1200, +0.0000)

(+0.8100, +0.0000)

(+0.8100, -0.0000)

(-0.2400, -0.0000)

(-0.2400, -0.0000)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (1,0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +6.7600, +18.9126, +27.4406, +20.1419, +3.9480, -4.1119, -2.2752, +0.2333, +0.3093, -0.0137, -0.0183, +0.0022

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{b}_T, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{a}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{a}_T \cdot \mathbf{b}_T, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{b}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{a}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{b}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{a}_T \cdot \mathbf{b}_T, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$

$(\mathbf{a}_T \cdot \mathbf{c}_T, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

(+1.1200, +0.0000)

(+1.1200, +0.0000)

(+1.1200, +0.0000)

(+1.1200, -0.0000)

(+0.8100, -0.0000)

(+0.8100, -0.0000)

(+0.8100, +0.0000)

(+0.8100, +0.0000)

... (4 more eval-pairs)

3.55 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1,0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +6.7600, +0.0000, +28.9706, +0.0000, +81.9304, -0.0000, +182.5098, -0.0000, +317.3109, -0.0000, +471.8350, -0.0000, +574.4634, +0.0000, +612.5140, +0.0000, +514.8609, +0.0000, +384.0874, +0.0000, +198.4450, +0.0000, +93.3348, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{b}_T, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$

$(\mathbf{a}_T \cdot \mathbf{b}_T, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{a}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{a}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{a}_T \cdot \mathbf{b}_T, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{a}_T \cdot \mathbf{b}_T, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
 $(+1.1200, -1.5900)$
 $(+0.8100, -1.5900)$
 $(-0.2400, -1.2000)$
 $(-0.2400, -1.0300)$
 $(+1.1200, -1.2000)$
 $(+0.8100, -1.0300)$
 $(+1.1200, +1.5900)$
 $(+0.8100, +1.5900)$
... (4 more eval-pairs)
PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (2, 0), type=12, kind=ASSOC, encoded dim= 12
Encoded values: +6.7600, +28.9706, +88.4413, +218.6210, +440.5627, +747.9918, +1056.1305, +1254.6568, +1207.9684, +922.4338, +489.6353, +157.4689
Atom-pair orbit (multiset; ordering arbitrary):
 $(\mathbf{a}_T \cdot \mathbf{b}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{a}_T \cdot \mathbf{b}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\mathbf{a}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{a}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\mathbf{a}_T \cdot \mathbf{b}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\mathbf{a}_T \cdot \mathbf{b}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
 $(-0.2400, -1.5900)$
 $(-0.2400, -1.5900)$
 $(+1.1200, -1.0300)$
 $(+1.1200, -1.0300)$
 $(+0.8100, -1.2000)$
 $(+0.8100, -1.2000)$
 $(-0.2400, +1.5900)$
 $(-0.2400, +1.5900)$
... (4 more eval-pairs)

3.56 Block: $\mathbf{x}_T \cdot \mathbf{y}_T \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (2, 0), type=12, kind=ASSOC, encoded dim= 6

Encoded values: +3.3800, +4.9844, +3.8602, +2.3357, +1.5450, +0.8403

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a}_T \cdot \mathbf{b}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{a}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{a}_T \cdot \mathbf{b}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$

$(\mathbf{a}_T \cdot \mathbf{c}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\mathbf{b}_T \cdot \mathbf{c}_T, -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 Eval-pair multiset (u, v) :
 $(-0.2400, -0.6430)$
 $(+1.1200, -0.6430)$
 $(+0.8100, -0.6430)$
 $(+1.1200, +0.6430)$
 $(+0.8100, +0.6430)$
 $(-0.2400, +0.6430)$

3.57 Block: $\mathbf{x} \cdot \mathbf{y} \times \mathbf{x} \cdot \mathbf{y}$ (block)

PairFlavour overlap=(1, 0) type=11_sigma kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=11_sigma, kind=ASSOC, encoded dim= 6

Encoded values: +3.6770, +7.2066, +10.3451, +9.7461, +6.3330, +2.7526

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a} \cdot \mathbf{b}, \mathbf{a} \cdot \mathbf{c})$
 $(\mathbf{a} \cdot \mathbf{c}, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{a} \cdot \mathbf{b}, \mathbf{b} \cdot \mathbf{c})$
 $(\mathbf{b} \cdot \mathbf{c}, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{a} \cdot \mathbf{c}, \mathbf{b} \cdot \mathbf{c})$
 $(\mathbf{b} \cdot \mathbf{c}, \mathbf{a} \cdot \mathbf{c})$

Eval-pair multiset (u, v) :

$(+0.9800, -0.5900)$
 $(+0.9100, -0.5900)$
 $(+0.9800, +0.9100)$
 $(+0.9100, +0.9800)$
 $(-0.5900, +0.9800)$
 $(-0.5900, +0.9100)$

PairFlavour overlap=(2, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (2, 0), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a} \cdot \mathbf{b}, \mathbf{a} \cdot \mathbf{b})$
 $(\mathbf{a} \cdot \mathbf{c}, \mathbf{a} \cdot \mathbf{c})$
 $(\mathbf{b} \cdot \mathbf{c}, \mathbf{b} \cdot \mathbf{c})$

Eval-pair multiset (u, v) :

$(-0.5900, -0.5900)$
 $(+0.9100, +0.9100)$
 $(+0.9800, +0.9800)$

3.58 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (0, 0), type=12, kind=ASSOC, encoded dim= 6

Encoded values: +2.6000, +1.2434, -1.6329, -1.3182, +0.2350, +0.2768

Atom-pair orbit (multiset; ordering arbitrary):

$(\mathbf{a} \cdot \mathbf{b}, \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{c}, \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{c}, \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{b}, -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{a} \cdot \mathbf{c}, -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\mathbf{b} \cdot \mathbf{c}, -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$

Eval-pair multiset (u, v) :

(-0.5900, +0.0000)
(-0.5900, +0.0000)
(+0.9800, +0.0000)
(+0.9800, -0.0000)
(+0.9100, -0.0000)
(+0.9100, -0.0000)

PairFlavour overlap=(1, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (1, 0), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +5.2000, +9.2468, +3.1999, -9.5814, -10.4452, +1.1640, +6.3288, +1.6586, -1.5236, -0.6746, +0.1301, +0.0766

Atom-pair orbit (multiset; ordering arbitrary):

(**a** · **b**, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
(**a** · **c**, $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
(**a** · **b**, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
(**b** · **c**, $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$)
(**a** · **c**, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
(**b** · **c**, $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$)
(**a** · **b**, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)
(**a** · **c**, $-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

(-0.5900, +0.0000)
(-0.5900, +0.0000)
(-0.5900, +0.0000)
(-0.5900, -0.0000)
(+0.9800, -0.0000)
(+0.9800, -0.0000)
(+0.9800, +0.0000)
(+0.9800, +0.0000)

... (4 more eval-pairs)

3.59 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +5.2000, +0.0000, +19.3048, -0.0000, +44.1431, -0.0000, +88.3113, -0.0000, +137.8122, -0.0000, +217.1767, -0.0000, +269.5218, -0.0000, +324.6213, -0.0000, +280.9677, -0.0000, +279.5834, -0.0000, +185.4809, -0.0000, +133.7534, -0.0000

Atom-pair orbit (multiset; ordering arbitrary):

(**a** · **b**, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
(**a** · **b**, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
(**a** · **c**, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
(**a** · **c**, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
(**a** · **b**, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
(**a** · **b**, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
(**b** · **c**, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
(**b** · **c**, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)

... (4 more atom-pairs)

Eval-pair multiset (u, v) :

(-0.5900, +1.2000)
(-0.5900, +1.0300)
(+0.9800, +1.5900)

(+0.9100, +1.5900)
 (+0.9800, +1.2000)
 (+0.9100, +1.0300)
 (−0.5900, −1.2000)
 (−0.5900, −1.0300)
 ... (4 more eval-pairs)
 PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (2, 0), type=12,
 kind=ASSOC, encoded dim= 12
Encoded values: +5.2000, +19.3048, +52.0675, +125.1057, +249.4131, +435.0611, +636.0779,
 +824.2870, +888.3968, +786.4776, +473.5668, +173.8698
 Atom-pair orbit (multiset; ordering arbitrary):
 (**a** · **b**, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 (**a** · **b**, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 (**a** · **c**, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$)
 (**a** · **c**, $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$)
 (**b** · **c**, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$)
 (**b** · **c**, $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$)
 (**a** · **b**, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$)
 (**a** · **b**, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$)
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v):
 (−0.5900, −1.5900)
 (−0.5900, −1.5900)
 (−0.5900, +1.5900)
 (−0.5900, +1.5900)
 (+0.9100, −1.2000)
 (+0.9100, −1.2000)
 (+0.9800, −1.0300)
 (+0.9800, −1.0300)
 ... (4 more eval-pairs)

3.60 Block: $\mathbf{x} \cdot \mathbf{y} \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=12 kind=ASSOC dim=6 **PairFlavour** overlap= (2, 0), type=12,
 kind=ASSOC, encoded dim= 6
Encoded values: +2.6000, +2.4837, +0.5170, +0.5921, +1.7446, +1.2990
 Atom-pair orbit (multiset; ordering arbitrary):
 (**a** · **b**, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 (**a** · **c**, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 (**b** · **c**, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 (**a** · **b**, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 (**a** · **c**, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 (**b** · **c**, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 Eval-pair multiset (u, v):
 (−0.5900, −0.6430)
 (−0.5900, +0.6430)
 (+0.9800, −0.6430)
 (+0.9100, −0.6430)
 (+0.9800, +0.6430)
 (+0.9100, +0.6430)

3.61 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 2) type=neg_sigma kind=ASSOC dim=6 **PairFlavour** overlap= (0, 2), type=neg_sigma, kind=ASSOC, encoded dim= 6

Encoded values: +0.0000, +0.0000, +0.0000, +0.0000, +0.0000, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
... (4 more atom-pairs)

Eval-pair multiset (u, v) :

(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(-0.0000, -0.0000)
(-0.0000, -0.0000)
(-0.0000, -0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
... (4 more eval-pairs)

PairFlavour overlap=(1, 2) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (1, 2), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

$(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}))$

Eval-pair multiset (u, v) :

(+0.0000, +0.0000)
(+0.0000, +0.0000)
(+0.0000, +0.0000)
(-0.0000, -0.0000)
(-0.0000, -0.0000)
(-0.0000, -0.0000)

3.62 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(0, 1) type=12 kind=ASSOC dim=12 **PairFlavour** overlap= (0, 1), type=12, kind=ASSOC, encoded dim= 12

Encoded values: +0.0000, +10.0580, +0.0000, +40.9913, +0.0000, +86.6819, +0.0000, +100.4717, +0.0000, +60.6378, +0.0000, +14.9163

Atom-pair orbit (multiset; ordering arbitrary):

$(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$

$(\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
 $(+0.0000, -1.5900)$
 $(+0.0000, -1.5900)$
 $(+0.0000, -1.2000)$
 $(-0.0000, -1.2000)$
 $(-0.0000, -1.0300)$
 $(-0.0000, -1.0300)$
 $(+0.0000, +1.5900)$
 $(+0.0000, +1.5900)$
... (4 more eval-pairs)
PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 1), type=11,
kind=ASSOC, encoded dim= 24
Encoded values: +0.0000, +0.0000, +10.0580, +0.0000, +0.0000, +0.0000, +40.9913, +0.0000,
+0.0000, +0.0000, +86.6819, +0.0000, +0.0000, +0.0000, +100.4717, +0.0000, +0.0000, +0.0000,
+60.6378, +0.0000, +0.0000, +0.0000, +14.9163, +0.0000
Atom-pair orbit (multiset; ordering arbitrary):
 $(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
 $(+0.0000, -1.5900)$
 $(+0.0000, -1.5900)$
 $(+0.0000, -1.2000)$
 $(-0.0000, -1.2000)$
 $(-0.0000, -1.0300)$
 $(-0.0000, -1.0300)$
 $(+0.0000, +1.5900)$
 $(+0.0000, +1.5900)$
... (4 more eval-pairs)

3.63 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=22 kind=ASSOC dim=6 **PairFlavour** overlap= (1, 0), type=22,
kind=ASSOC, encoded dim= 6

Encoded values: -2.4807, +2.5641, -1.4135, +0.4383, -0.0725, +0.0050

Atom-pair orbit (multiset; ordering arbitrary):

$(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$

$(\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 $(-\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c}))$
 ... (4 more atom-pairs)
 Eval-pair multiset (u, v) :
 $(+0.0000, +0.6430)$
 $(+0.0000, +0.6430)$
 $(+0.0000, -0.6430)$
 $(-0.0000, -0.6430)$
 $(-0.0000, +0.6430)$
 $(-0.0000, +0.6430)$
 $(+0.0000, -0.6430)$
 $(+0.0000, -0.6430)$
 ... (4 more eval-pairs)

3.64 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(1, 0) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 0), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +0.0000, +0.0000, +0.0000, +9.2388, -0.0000, -0.0000, -14.7572, +0.0000, +0.0000, -0.0000, +0.0000, +98.5481, -0.0000, +0.0000, -301.4282, +0.0000, +0.0000, -0.0000, -0.0000, +257.2546, +0.0000, +0.0000, -1268.5418, -0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 ... (4 more atom-pairs)

Eval-pair multiset (u, v) :

$(-1.0300, +1.5900)$
 $(-1.2000, +1.5900)$
 $(-1.5900, +1.2000)$
 $(-1.5900, +1.0300)$
 $(+1.0300, +1.2000)$
 $(+1.2000, +1.0300)$
 $(-1.2000, -1.0300)$
 $(-1.0300, -1.2000)$

... (4 more eval-pairs)

PairFlavour overlap=(1, 1) type=11 kind=ASSOC dim=24 **PairFlavour** overlap= (1, 1), type=11, kind=ASSOC, encoded dim= 24

Encoded values: +0.0000, +0.0000, +0.0000, -9.2388, +0.0000, -0.0000, -14.7572, -0.0000, +0.0000, +0.0000, -0.0000, -98.5481, +0.0000, -0.0000, -301.4282, -0.0000, -0.0000, -0.0000, -0.0000, -257.2546, +0.0000, +0.0000, -1268.5418, +0.0000

Atom-pair orbit (multiset; ordering arbitrary):

$(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$

$(\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
(+1.5900, +1.0300)
(+1.5900, +1.2000)
(+1.2000, +1.5900)
(+1.0300, +1.5900)
(+1.2000, -1.0300)
(+1.0300, -1.2000)
(-1.0300, +1.2000)
(-1.2000, +1.0300)
... (4 more eval-pairs)
PairFlavour overlap=(2, 0) type=neg_sigma kind=ASSOC dim=6 **PairFlavour** overlap= (2, 0), type=neg_sigma,
kind=ASSOC, encoded dim= 6
Encoded values: -20.1160, -163.9651, +693.4549, +1607.5466, -1940.4080, -954.6460
Atom-pair orbit (multiset; ordering arbitrary):
 $(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}), -\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
(+1.5900, -1.5900)
(-1.5900, +1.5900)
(+1.5900, -1.5900)
(-1.5900, +1.5900)
(+1.2000, -1.2000)
(-1.2000, +1.2000)
(+1.2000, -1.2000)
(-1.2000, +1.2000)
... (4 more eval-pairs)
PairFlavour overlap=(2, 1) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (2, 1),
type=null_self, kind=NULL_SELF, encoded dim= 0
Atom-pair orbit (multiset; ordering arbitrary):
 $(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
 $(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p}))$
 $(\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}), \varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q}))$
 $(\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p}))$
 $(\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}), \varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p}))$
 $(-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}), -\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q}))$
... (4 more atom-pairs)
Eval-pair multiset (u, v) :
(+1.0300, +1.0300)
(+1.0300, +1.0300)
(+1.2000, +1.2000)
(+1.2000, +1.2000)

(+1.5900, +1.5900)
 (+1.5900, +1.5900)
 (−1.0300, −1.0300)
 (−1.0300, −1.0300)
 ... (4 more eval-pairs)

3.65 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(2, 0) type=neg kind=ASSOC dim=12 **PairFlavour** overlap= (2, 0), type=neg, kind=ASSOC, encoded dim= 12

Encoded values: +7.5773, +0.0000, +31.0800, -0.0000, +72.8464, -0.0000, +116.6920, -0.0000, +108.6872, -0.0000, +64.6121, -0.0000

Atom-pair orbit (multiset; ordering arbitrary):

($\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ... (4 more atom-pairs)

Eval-pair multiset (u, v):

(+1.5900, −0.6430)
 (−1.5900, +0.6430)
 (+1.5900, +0.6430)
 (−1.5900, −0.6430)
 (+1.2000, −0.6430)
 (−1.2000, +0.6430)
 (+1.0300, −0.6430)
 (−1.0300, +0.6430)
 ... (4 more eval-pairs)

3.66 Block: $\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z}) \times \varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (block)

PairFlavour overlap=(3, 0) type=null_self kind=NULL_SELF dim=0 **PairFlavour** overlap= (3, 0), type=null_self, kind=NULL_SELF, encoded dim= 0

Atom-pair orbit (multiset; ordering arbitrary):

($\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$, $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)
 ($-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$, $-\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$)

Eval-pair multiset (u, v):

(+0.6430, +0.6430)
 (−0.6430, −0.6430)

4 Alignment Decoder (recovers the G-orbit of repS evaluations)

—repS—: $|\text{repS}| = 11$

—G—: $|G| = 12$

Decoded vectors: decoded column vectors: 12

Decoded alignment table (rows=repS atoms, cols=group elements): **Decoded alignment table**
(rows= repS atoms, columns= $g \in G$):

Table 1: Decoded alignment table

Atom	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	
$ \mathbf{a} $	+1.4765	+1.4765	+1.3342	+1.3342	+1.1874	+1.1874	+1.1874	+1.1874	+1.3342	+1.3342
$ \mathbf{p} $	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000
$\mathbf{a} \cdot \mathbf{b}$	-0.5900	+0.9800	-0.5900	+0.9100	+0.9800	+0.9100	+0.9100	+0.9800	+0.9100	-0.5900
$\mathbf{a} \cdot \mathbf{p}$	-0.7000	-0.7000	-0.5000	-0.5000	-0.2000	-0.2000	+0.2000	+0.2000	+0.5000	+0.5000
$\mathbf{p} \cdot \mathbf{q}$	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000
$\mathbf{a}_T \cdot \mathbf{b}_T$	-0.2400	+1.1200	-0.2400	+0.8100	+1.1200	+0.8100	+0.8100	+1.1200	+0.8100	-0.2400
$\mathbf{a}_T \cdot \mathbf{p}_T$	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000
$\mathbf{p}_T \cdot \mathbf{q}_T$	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430	-0.6430
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$	-1.5900	-1.0300	-1.5900	-1.2000	-1.0300	+1.2000	-1.2000	+1.0300	+1.2000	+1.5900
$\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$	+0.0000	+0.0000	+0.0000	+0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000	-0.0000

Ground truth (direct evaluation of g-repS on event): **Ground truth** (direct evaluation of $g \cdot \text{repS}$ on event):

Table 2: Ground truth alignment table

Atom	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	
$ \mathbf{a} $	+1.3342	+1.3342	+1.3342	+1.3342	+1.1874	+1.1874	+1.1874	+1.1874	+1.4765	+1.4765
$ \mathbf{p} $	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000
$\mathbf{a} \cdot \mathbf{b}$	+0.9100	+0.9100	-0.5900	-0.5900	+0.9100	+0.9100	+0.9800	+0.9800	-0.5900	-0.5900
$\mathbf{a} \cdot \mathbf{p}$	+0.5000	-0.5000	+0.5000	-0.5000	+0.2000	-0.2000	+0.2000	-0.2000	-0.7000	+0.7000
$\mathbf{p} \cdot \mathbf{q}$	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000
$\mathbf{a}_T \cdot \mathbf{b}_T$	+0.8100	+0.8100	-0.2400	-0.2400	+0.8100	+0.8100	+1.1200	+1.1200	-0.2400	-0.2400
$\mathbf{a}_T \cdot \mathbf{p}_T$	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000
$\mathbf{p}_T \cdot \mathbf{q}_T$	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000	+0.0000
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$	-0.6430	-0.6430	+0.6430	+0.6430	+0.6430	+0.6430	-0.6430	-0.6430	-0.6430	-0.6430
$\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$	+1.2000	-1.2000	+1.5900	-1.5900	-1.2000	+1.2000	+1.0300	-1.0300	-1.5900	+1.5900
$\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000

Multiset comparison (decoded vs ground truth): **Multiset comparison:**

MISMATCH — decoded multiset differs from ground truth

Max —error— per repS row (decoded vs ground truth): **Max |error| per repS row:**

Mymag(a): $|\mathbf{a}|$: max err = 0.00e + 00

Mymag(p): $|\mathbf{p}|$: max err = 0.00e + 00

dot(a,b): $\mathbf{a} \cdot \mathbf{b}$: max err = 0.00e + 00

dot(a,p): $\mathbf{a} \cdot \mathbf{p}$: max err = 0.00e + 00

dot(p,q): $\mathbf{p} \cdot \mathbf{q}$: max err = 0.00e + 00

dot(a,b): $\mathbf{a}_T \cdot \mathbf{b}_T$: max err = $0.00e + 00$

dot(a,p): $\mathbf{a}_T \cdot \mathbf{p}_T$: max err = $0.00e + 00$

dot(p,q): $\mathbf{p}_T \cdot \mathbf{q}_T$: max err = $0.00e + 00$

eps(a,b,c): $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$: max err = $1.29e + 00$

eps(a,b,p): $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$: max err = $0.00e + 00$

eps(a,p,q): $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$: max err = $0.00e + 00$

5 Summary

Phase 1 encoded length: Phase 1 encoded length: 35 reals

Phase 2 encoded length: Phase 2 encoded length: 1214 reals

Total encoded length: Total encoded length: 1249 reals

repS size: $|\text{repS}| = 11$ atoms

—G—: $|G| = 12$ group elements

Decoded vectors: decoded alignment vectors: 12

TeX output written to: /Users/lester/github/Echinocoder/symcoder/demo_roundtrip.tex Compile with: pdflatex demo_roundtrip.tex